Aalto Robotic Club Presents

ROS2 WORKSHOP

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Aalto University

Slides by Risto Ojala, based on previous version by Jari Vepsäläinen & Pavithra Gopalakrishnan
Now Presented by Pejman and Nick…

Contents

Robotic Operating System

What is ROS? Why is it used?

The fundamentals of ROS: packages, nodes, topics, messages

Docker

What is a container and why we like them? Why we hate them?

Some Hands-on Experience

Lets play around with ROS And understand it better.

Open Discussion

Disclaimer!

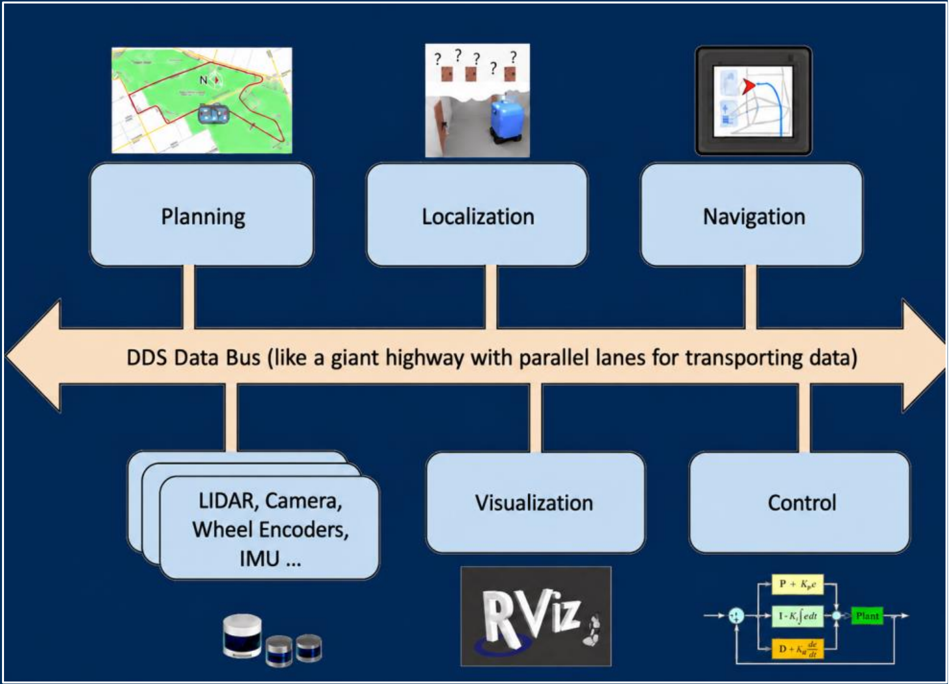
Throughout the workshop, the term ROS is used to refer to ROS 2! Traditionally ROS has been used to refer to ROS1 However, ROS1 has reached end of life, and is now only used in legacy systems.

What is ROS?

ROS = Robot Operating System

ROS is a middleware solution for robotics software, with some convenient tools for development

ROS enables modular software implementation, handling communication between different software components.



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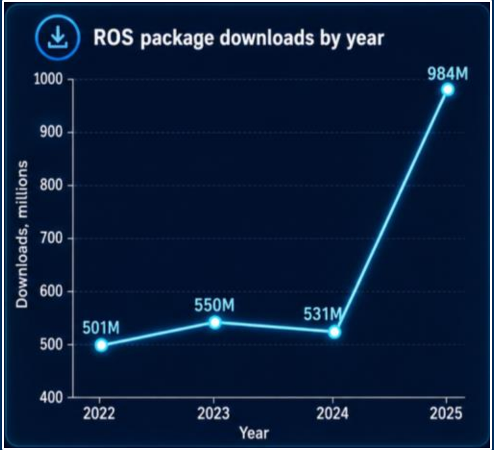
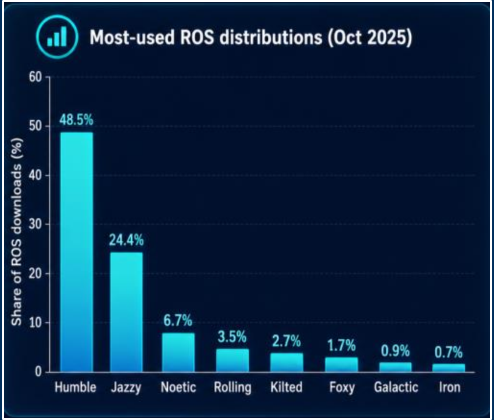
Why ROS?





Modularity

Development can focus on individual robotic functionalities, “the big-picture” (the communication framework) is ready-made Convenient integration of individual software components

Community

Initial release of ROS was in 2007, development continues actively
Extensive community with a large number of open-source packages available

Convenient developer tools

Rviz & rqt (visualisation), rosbags (data recording)...

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Let's Speak ROS!

Package

ROS-based software implementation. Contains a ROS-based software entity which can be installed / shared

Node

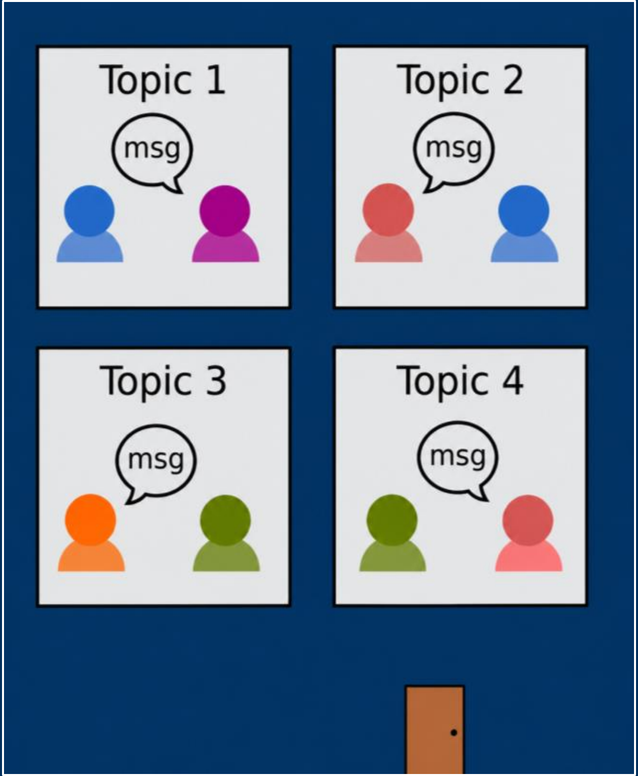
Basic unit in a ROS graph, running an implemented functionality

Topic

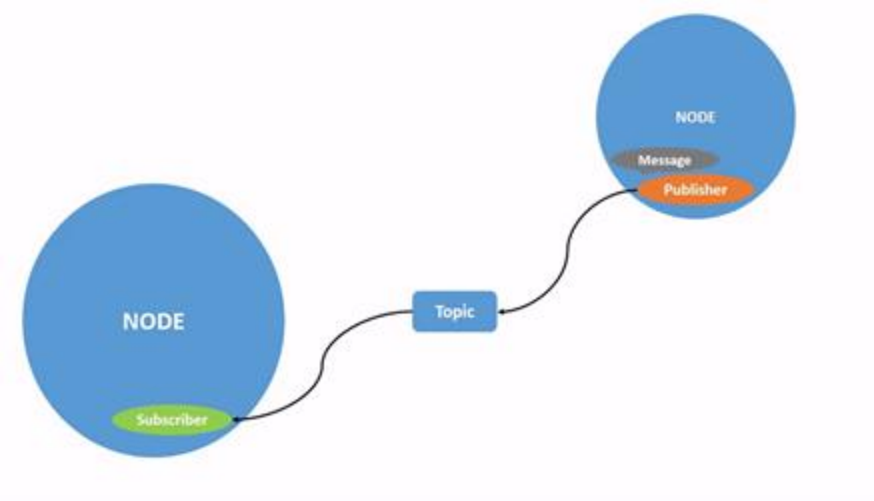
A continuous data stream, used by nodes to exchange information

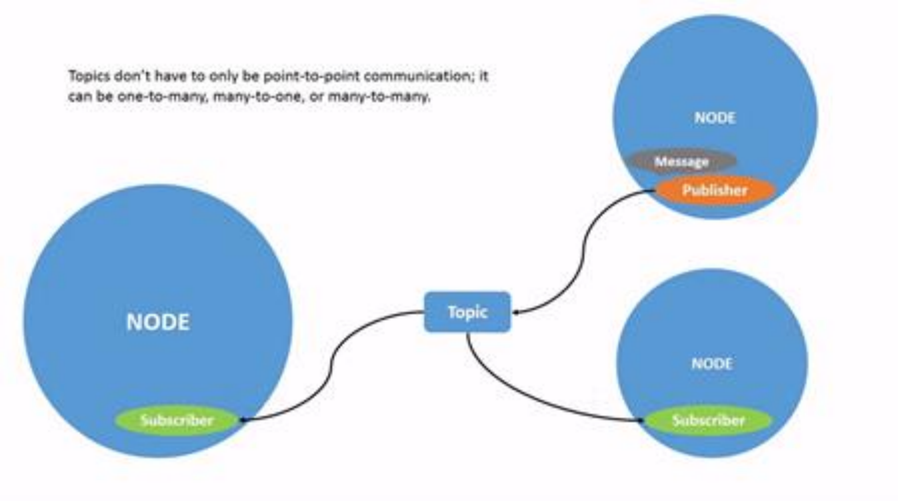
Message

Information in predefined format, passed within a topic (published /subscribed)



Pub Sub System





Workshop Starts!

Docker Container

An open-source platform that automates the deployment of applications inside lightweight, portable, and self-sufficient containers.

Why is it useful?

ROS often requires strict compatibility between ROS versions, Ubuntu versions, embedded systems, dependencies, and development environments. Docker skips the whole process.

Why might it not be ideal?

Docker can add complexity when working with hardware access, real-time performance, graphical interfaces, networking, or embedded devices.

How are we using Docker today?

We would download and image, run it and use visualization tools through WebSocket.

Download Docker and the image!

docker pull pjmrobotics/ros2-turtlebot3-vnc:latest

```
docker run --rm -it
--name ros2-turtlebot3-vnc
--shm-size=2g
-p 6080:80
-p 5900:5900
pjmrobotics/ros2-turtlebot3-vnc:latest
```

Gazebo

```
export TURTLEBOT3_MODEL=burger
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

RVIZ

```
rviz2
```

Drive around!

```
ros2 run turtlebot3_teleop teleop_keyboard
```

Open Discussion

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Contact info

PJ LinkedIn

ARC tg chat

Quiz Time

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